

Transfer Learning for Forest Image Classification Using EfficientNet-B2

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Abstract—Forest management has become more challenging because of increasing fire risk, drought, and vegetation stress. Manual forest management is time-consuming and difficult to apply across large areas. This study was developed as part of the Sky Innovators Project, which aims to support environmental monitoring and reforestation planning through AI-assisted visual analysis. Three image classification modules were developed and evaluated: forest/non-forest classification, forest density estimation, and fire/no-fire classification. The image datasets were manually reviewed and cleaned before training. All three modules used EfficientNet-B2 with ImageNet pretrained weights. All three models were trained using AdamW and Focal Loss. The forest/non-forest model achieved 99.73% test accuracy and a macro F1-score of 0.9954. The density model achieved 93.04% test accuracy and a macro F1-score of 0.9026. The fire/no-fire model achieved 99.80% test accuracy and a macro F1-score of 0.9922. Since all test sets were drawn from the same main data distributions, these results should be interpreted as internal performance on the prepared datasets rather than as evidence of generalization to external drone, aerial, or satellite imagery. Future work will test the models on independent external datasets.

Index Terms—forest/non-forest classification, forest density estimation, fire/no-fire classification, EfficientNet-B2, transfer learning, remote sensing imagery, Focal Loss, class imbalance, environmental monitoring, forest management

I. INTRODUCTION

A. Problem Statement and Motivation

Forests are large natural ecosystems primarily covered by trees and vegetation that provide essential ecological, environmental, and climatic functions. According to the Food and Agriculture Organization (FAO), forests cover approximately 31% of the Earth's total land area worldwide [1]. Forests also function as major carbon sinks by absorbing large amounts of atmospheric carbon dioxide (CO_2) through photosynthesis [2]. The United Nations Environment Programme (UNEP) reports that forests absorb approximately 2.6 billion tons of CO_2 annually, making them highly important for mitigating climate change and reducing greenhouse gas accumulation [2]. Despite their ecological importance, forests are increasingly threatened by wildfires, drought, and deforestation. Wildfires caused approximately 160 million hectares (Mha) of tree cover loss worldwide between 2001 and 2025, accounting for nearly 30% of total global tree cover loss during this period [3]. Drought conditions are among the major factors increasing wildfire frequency and severity worldwide [4]. Deforestation

also remains a major environmental challenge, with approximately 10.9 million hectares of forest land lost annually according to FAO reports [1]. Therefore, forest conservation, sustainable management, and ecosystem restoration are considered critical strategies for mitigating climate change and maintaining global ecological stability [1]. Although forests are under high risk, traditional forest monitoring methods often rely on manual field surveys and expert interpretation, which are time-consuming, labor-intensive, and difficult to scale across large geographic areas [5]. As UAV, aerial, and satellite imaging technologies become increasingly accessible, remote sensing-based image classification has emerged as an effective alternative for large-scale environmental monitoring [6]. Aerial and satellite imagery can capture extensive forest regions rapidly and repeatedly, enabling continuous observation of wildfire risks, vegetation density, drought effects, and land-cover changes [6]. Traditional machine learning methods were initially used for remote sensing image classification; however, these approaches often rely on handcrafted feature extraction techniques, which may have limited generalization capability across different environmental conditions and image acquisition settings [7]. In contrast, deep learning-based approaches can automatically learn hierarchical spatial and textural representations directly from image data, enabling more robust classification performance in remote sensing applications [8]. In particular, convolutional neural network (CNN)-based architectures have demonstrated strong performance in land-cover classification, vegetation analysis, and aerial scene recognition tasks [8]. Architectures such as ResNet, DenseNet, MobileNet, and EfficientNet have therefore become widely adopted in environmental image classification studies due to their ability to efficiently extract complex visual patterns from high-dimensional imagery [8]. Many existing remote sensing studies focus on single-task classification using relatively homogeneous benchmark datasets collected from specific sensors or acquisition conditions. Limited attention is given to iterative dataset cleaning, heterogeneous image sources, and the effect of ambiguous or visually unclear samples on model reliability. In addition, several studies report high classification accuracy without extensively discussing dataset generalization across different regions, seasons, sensors, or acquisition settings. These limitations indicate the need for more application-oriented forest monitoring studies that combine multiple classification tasks with carefully refined datasets and heterogeneous environmental imagery. This

study investigates whether EfficientNet-based transfer learning models can effectively support multiple forest monitoring tasks using heterogeneous RGB image datasets collected from aerial, drone, satellite, and publicly available image sources. The work focuses on three related classification problems: forest/non-forest classification, forest density estimation, and fire/no-fire classification. In addition to model performance, the study also examines the role of iterative dataset cleaning and manual label refinement in improving classification reliability for forest-related environmental monitoring tasks. The main contributions of this study are summarized as follows:

- The study investigates the role of iterative dataset refinement and manual label correction in improving reliability for forest-related image classification tasks.
- A heterogeneous forest/non-forest classification dataset was assembled from aerial, drone, satellite, and publicly available RGB image sources.
- EfficientNet-B2-based transfer learning models were developed and evaluated for three related forest monitoring tasks: forest/non-forest classification, forest density estimation, and fire/no-fire classification.
- The study analyzes the challenges of visually ambiguous environmental imagery, including class overlap, label noise, and varying acquisition conditions across multiple forest monitoring tasks.
- The limitations of using internal test sets from similar source distributions were explicitly discussed to provide a more transparent interpretation of model generalization performance.

II. RELATED WORK

Deep learning has become a common approach in remote sensing image classification [8]. Convolutional neural networks can learn spatial and textural patterns directly from image data [8]. Architectures such as ResNet, VGG, DenseNet, MobileNet, and EfficientNet have therefore been used in land-cover classification, scene recognition, and vegetation classification tasks [8].

A. Forest/Non-Forest Classification

Recent work has investigated deep learning-based forest and vegetation classification using remote sensing imagery. Jain et al. proposed an EfficientNet-B0-based model for forest/non-forest classification using Sentinel-2 satellite imagery [9]. Their model used ImageNet pretrained weights and combined RGB bands with the Normalized Difference Vegetation Index (NDVI) as a fourth input channel [9]. The study reported 98% accuracy and a 98% F1-score, suggesting that EfficientNet-based models can provide strong performance while using fewer parameters than larger CNN architectures such as VGG-16 and ResNet-50 [9]. The authors also emphasized the importance of class balance and diverse non-forest examples for improving generalization [9]. Unlike the multispectral Sentinel-2 imagery used by Jain et al., the present study focuses on heterogeneous RGB imagery collected from multiple aerial, drone, satellite, and publicly available image sources.

B. Fire/No-Fire Classification

Early wildfire detection is important for reducing ecological damage and improving emergency response efficiency [10]. Recent studies have shown that deep learning-based fire detection systems can support automated real-time wildfire monitoring using surveillance and aerial imagery [10]. CNN-based models are particularly suitable for fire/no-fire classification because they can automatically learn discriminative visual patterns such as flame texture, smoke appearance, and illumination variations from image data [10]. However, wildfire detection remains challenging due to visual similarities between smoke, haze, clouds, sunlight reflections, and bright non-fire regions [10]. Muhammad et al. noted that conventional fire detection systems often struggle in large and complex outdoor environments and may fail to detect fires during early stages [10]. Their study proposed a CNN-based fire detection framework for surveillance applications and achieved 96.8% classification accuracy [10].

C. Forest Density Classification

Forest density classification has also been studied in drone-based image analysis. Khidhir and Sehree [11] proposed a CNN-based approach for classifying tree density into low, medium, and high categories using UAV images. Their study compared VGG19 and ResNet-50 architectures and reported 84.07% accuracy for VGG19 and 91.17% accuracy for ResNet-50. The authors noted that larger datasets and additional training could improve classification performance, especially for visually similar density levels. While Khidhir and Sehree focused specifically on UAV imagery for density estimation, the present study combines heterogeneous image sources and additionally investigates the effect of iterative dataset cleaning on classification reliability.

D. Dataset Quality and Generalization Challenges

Existing studies on forest and land-cover classification often focus on satellite or multispectral imagery from established remote sensing datasets. These studies often show that deep learning models can outperform traditional machine learning methods when sufficient labeled data is available. However, dataset quality remains a key challenge. Many studies use public datasets directly and pay limited attention to manual cleaning, label correction, or the removal of ambiguous samples. In real-world image classification tasks, noisy labels, ambiguous samples, and visually unclear images can reduce model reliability. For instance, González-Santoyo et al. applied the Cumulative Spectral Gradient (CSG) method to mitigate label noise and improve model accuracy in image classification tasks [12]. Their study used noisy CIFAR-10N and CIFAR-100N datasets containing labels with different levels of noise [12]. DenseNet and Xception architectures were employed to analyze the effects of label correction on model performance [12]. While the worst-case dataset containing 40% label noise achieved 66% accuracy, the clean dataset achieved 94% accuracy [12]. These results demonstrate that

label correction can significantly improve model precision and classification reliability [12].

Similarly, Northcutt et al. proposed the Confident Learning approach for identifying and characterizing label errors in datasets [13]. Their study investigated the effect of removing label issues from ImageNet training data and retraining ResNet-18 models from scratch [13]. The results showed that Confident Learning achieved better training performance compared to random sample removal strategies [13]. These findings indicate that dataset preparation, label correction, and iterative cleaning are critical components of reliable image classification systems. Therefore, this study emphasizes dataset cleaning and manual label refinement as important parts of the classification pipeline.

E. Research Gap and Motivation

One limitation in some studies is that high accuracy is reported without clearly stating whether the test data comes from the same source distribution as the training data or from an independent external dataset. This distinction is important because a model may perform well on visually similar test images but perform worse on images collected from different regions, seasons, sensors, or acquisition conditions. Therefore, this study discusses the limitations of using internal test sets for all three modules.

While many land-cover classification studies focus on multi-class problems, forest/non-forest classification is simpler than multi-class land-cover classification and it remains practically useful for identifying forested areas at scale. The density and fire/no-fire modules add finer-grained forest characterization and environmental risk detection, which are relevant for reforestation planning and environmental monitoring. Two widely used benchmark datasets for remote sensing and aerial image classification are EuroSAT and AID [14], [15]. EuroSAT is a satellite-based land-use dataset derived from Sentinel-2 imagery and contains 27,000 images across 10 land-use classes [14]. AID includes 10,000 aerial images across 30 scene categories and is widely used for evaluating aerial scene classification models [15].

This study differs from these benchmark datasets in several ways. First, it addresses three separate classification tasks rather than general multi-class scene recognition: forest/non-forest classification, forest density estimation, and fire/no-fire classification. Second, the datasets were assembled from heterogeneous sources, including drone, aerial, satellite, and other publicly available image sources, rather than from a single sensor or benchmark. Third, the images were manually labeled and refined through iterative cleaning. In this way, the study builds on previous work in remote sensing image classification and gives more attention to dataset quality and application-specific classification tasks.

III. METHODOLOGY

The methodology consists of three classification modules. All three modules use the same model architecture, image

resolution, and batch size. The following sections describe the dataset preparation and training configuration for each module.

A. Module 1: Forest/Non-Forest Classification

1) *Dataset Preparation*: The dataset was prepared as a classification dataset with two categories:

- **Forest**: dense forest areas, sparse tree-covered regions, shrubland-like vegetation, and vegetated areas relevant to drought monitoring or reforestation.
- **Non-forest**: urban areas, roads, buildings, agricultural fields, water bodies, bare land, and other non-forested regions.

In total, 34,678 images were manually reviewed and labeled. Images containing tree cover, shrubland-like vegetation, or vegetation relevant to drought monitoring and reforestation were assigned to the forest class.



Fig. 1. Representative examples from the forest and non-forest classes. Left: forest. Right: non-forest.

2) *Iterative Dataset Cleaning*: The forest/non-forest dataset was refined through several experimental iterations. In the early experiments, the model achieved lower accuracy, which motivated a closer review of the dataset and model outputs. During this process, low-quality, ambiguous, visually similar, and incorrectly labeled images were removed or corrected. Samples that did not clearly represent either the forest or non-forest class were also excluded. New higher-quality images were added to improve dataset diversity and quality. After each dataset update, the revised dataset was re-uploaded and the train/validation/test split was recreated.

3) *Data Splitting and Class Imbalance Handling*: The final dataset contained 34,678 images, including 6,980 forest images and 27,698 non-forest images, with an approximate 1:4 class imbalance. To reduce the effect of this imbalance, Focal Loss [16] was used during training, and weighted random sampling was applied at the batch level to oversample the minority forest class. Table I shows the dataset split.

4) *Training Configuration*: The forest/non-forest model used AdamW as the optimizer and Focal Loss as the loss function. The primary checkpoint selection metric was the validation F1-score for the forest class. Training was performed for 30 epochs in two stages using a Kaggle GPU environment. In Stage 1 (epochs 1–10), the EfficientNet-B2 backbone was frozen, and the classifier head was trained with a learning rate of 1×10^{-3} . In Stage 2 (epochs 11–30), the

TABLE I
FOREST/NON-FOREST CLASSIFICATION DATASET SPLIT

Set	Number of Images
Training	24,274
Validation	5,202
Test	5,202
Total	34,678

last three feature blocks and the classifier were unfrozen. A differential learning rate strategy was applied: 1×10^{-5} for the unfrozen feature blocks and 1×10^{-4} for the classifier layer. CosineAnnealingLR was used in both stages.

B. Module 2: Forest Density Classification

1) *Dataset Preparation*: The density dataset contained 6,027 images across three classes: `low_density`, `medium_density`, and `high_density`. These classes correspond to low, medium, and high forest density levels. The dataset split is shown in Table II. The density images were manually reviewed, and unclear or visually ambiguous samples were removed or corrected when necessary.



Fig. 2. Representative examples from the forest density classes. Left: low density. Middle: medium density. Right: high density.

TABLE II
DENSITY CLASSIFICATION DATASET SPLIT

Set	Number of Images
Training	4,218
Validation	904
Test	905
Total	6,027

2) *Training Configuration*: The density model used AdamW as the optimizer and Focal Loss with $\gamma = 2.0$ as the loss function. The primary checkpoint selection metric was the validation macro F1-score. Training was performed in three stages. The best checkpoint was obtained at epoch 35 in Stage 3. It achieved a validation macro F1-score of 0.9126, validation accuracy of 93.25%, and validation loss of 0.1338. The full training configuration is summarized in Table III.

TABLE III
DENSITY MODEL TRAINING CONFIGURATION

Component	Value
Model	EfficientNet-B2
Image size	260×260
Optimizer	AdamW
Loss function	Focal Loss ($\gamma = 2.0$)
Batch size	32
Training schedule	Three-stage fine-tuning
Primary metric	Validation Macro F1-score
Best epoch	35 (Stage 3)
Best val macro F1	0.9126
Best val accuracy	93.25%
Best val loss	0.1338

C. Module 3: Fire/No-Fire Classification

1) *Dataset Preparation*: The fire/no-fire dataset contained 6,601 images across two classes: `fire` and `no_fire`. The dataset split is shown in Table IV. The fire/no-fire images were manually reviewed, and unclear, duplicate, or visually misleading samples were removed when necessary.



Fig. 3. Representative examples from the fire and no-fire classes. Left: fire. Right: no-fire.

TABLE IV
FIRE/NO-FIRE CLASSIFICATION DATASET SPLIT

Set	Number of Images
Training	4,620
Validation	990
Test	991
Total	6,601

2) *Training Configuration*: The fire/no-fire model used AdamW as the optimizer and Focal Loss with $\gamma = 2.0$ as the loss function. The primary checkpoint selection metric was the validation F1-score for the fire class. In Stage 1, only the classifier head was trained, while Stage 2 further fine-tuned the model. The best checkpoint was obtained at epoch 15 in Stage 2. It achieved a validation fire F1-score of 1.0000, validation macro F1-score of 1.0000, validation accuracy of 1.0000, and validation loss of 0.0002. The full training configuration is summarized in Table V.

D. Shared Components Across All Three Modules

1) *Model Architecture*: All three modules used EfficientNet-B2 [17] with ImageNet pretrained weights [18].

TABLE V
FIRE/NO-FIRE MODEL TRAINING CONFIGURATION

Component	Value
Model	EfficientNet-B2
Image size	260 × 260
Optimizer	AdamW
Loss function	Focal Loss (gamma = 2.0)
Batch size	32
Primary metric	Validation Fire F1-score
Best epoch	15 (Stage 2)
Best val fire F1	1.0000
Best val macro F1	1.0000
Best val accuracy	1.0000
Best val loss	0.0002

EfficientNet-B2 scales depth, width, and input resolution in a balanced way and provides a good balance between classification performance and training cost. EfficientNet-B2 was chosen for several reasons. First, it uses compound scaling across depth, width, and resolution, which gives good classification performance without needing as many parameters as VGG-16 or ResNet-50. Second, Jain et al. [9] showed that EfficientNet-based models work well on forest classification tasks. Third, the 260×260 input resolution fits well with aerial and drone images where spatial detail matters. A comparison with other architectures is planned for future work. Each model was fine-tuned for its task using transfer learning.

2) *Data Augmentation*: Lightweight data augmentation was applied only to the training sets to make the model more robust and reduce the risk of overfitting [19]. The augmentation pipeline included random horizontal flipping ($p = 0.5$), random rotation up to $\pm 10^\circ$, and color jitter with brightness, contrast, and saturation factors of 0.15 and a hue factor of 0.03. No augmentation was applied to the validation or test sets. All images were normalized using ImageNet statistics: mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225].

3) *Data Source and Ethics Note*: The datasets were collected from Kaggle and other publicly available sources. During dataset preparation, images from different public sources were combined, manually reviewed, and cleaned through several experimental iterations. Since the dataset was updated multiple times during this process, the exact number of images retained from each original source was not recorded.

The final datasets are therefore reported at the module level rather than at the individual source level. The forest/non-forest module contained 34,678 manually reviewed images, the density module contained 6,027 manually reviewed images, and the fire/no-fire module contained 6,601 manually reviewed images.

The study used aerial, drone, satellite, and land-cover images rather than personal or sensitive data, so no personally identifiable information was used in the experiments.

IV. EXPERIMENTS AND RESULTS

A. Module 1: Forest/Non-Forest Classification

During development, multiple experiments were performed to improve both the training pipeline and the dataset quality. In the early experiments, the model achieved approximately 84% accuracy. After improvements in the training setup and dataset preparation, accuracy increased to approximately 92%. Further improvement was achieved by removing ambiguous and low-quality images, correcting mislabeled samples, and adding higher-quality images.

Table VI summarizes the main experimental iterations. In the final experiment, the EfficientNet-B2 model achieved the test performance reported in Table VII. The best checkpoint was obtained at epoch 29, with a validation F1-score of 0.9898 for the forest class.

The confusion matrix for the forest/non-forest model is shown in Table VIII. Among 5,202 test images, only one forest image was misclassified as non-forest, and 13 non-forest images were misclassified as forest.

B. Module 2: Forest Density Classification

The density model achieved the test performance reported in Table IX. The model performed strongly on *low_density* and *high_density* classes, while *medium_density* was more challenging due to visual overlap with adjacent density levels.

The confusion matrix for the density model is reported in Table X. The most frequent errors involved the *medium_density* class being confused with *low_density* or *high_density*, reflecting the visual ambiguity of intermediate density levels.

C. Module 3: Fire/No-Fire Classification

The fire/no-fire model achieved the test performance reported in Table XI. The *fire* class achieved an F1-score of 0.9855, while the *no_fire* class achieved an F1-score of 0.9989, indicating strong internal test performance for distinguishing visible fire patterns from no-fire images.

The confusion matrix for the fire/no-fire model is shown in Table XII. Among 991 test images, 68 *fire* images were correctly classified as *fire*, while one *fire* image was misclassified as *no_fire*. Similarly, 921 *no_fire* images were correctly classified, while one *no_fire* image was misclassified as *fire*.

D. Cross-Module Comparison

Table XIII provides a summary comparison of the three modules. Forest/non-forest and fire/no-fire models achieved the highest accuracy, which is expected given the clearer visual distinction between their class categories. The density module is more challenging because its categories are finer-grained and visually similar.

V. DISCUSSION

A. Forest/Non-Forest Classification

The forest/non-forest model achieved strong performance on the prepared test set, with training and validation curves showing no clear sign of overfitting. During Stage 2, the

TABLE VI
EFFECT OF ITERATIVE DATASET CLEANING ON FOREST/NON-FOREST CLASSIFICATION TEST PERFORMANCE

Exp.	Dataset Size	Primary Change	Best Epoch	Test Acc.	Macro F1	Forest F1	Non-Forest F1
1	~27,000	Initial dataset; baseline pipeline	—	~85%	—	—	—
2	~34,750	Improved pipeline; dataset expanded	—	~92%	—	—	—
3	34,374	Iterative cleaning; mislabeled images re-moved	16	99.22%	0.9866	0.9779	0.9953
Final	34,678	Full cleaning; higher-quality images added	29	99.73%	0.9954	0.9925	0.9984

TABLE VII
FOREST/NON-FOREST CLASSIFICATION: FINAL TEST RESULTS

Metric	Result
Test Loss	0.0042
Test Accuracy	99.73%
Test Macro F1-score	0.9954
Forest Precision	0.9862
Forest Recall	0.9989
Forest F1-score	0.9925
Non-Forest Precision	0.9998
Non-Forest Recall	0.9970
Non-Forest F1-score	0.9984

TABLE VIII
FOREST/NON-FOREST CLASSIFICATION: CONFUSION MATRIX ON THE FINAL TEST SET

True \ Predicted	Forest	Non-Forest
Forest	929	1
Non-Forest	13	4259

TABLE IX
DENSITY CLASSIFICATION: FINAL TEST RESULTS

Metric	Result
Test Loss	0.0927
Test Accuracy	93.04%
Test Macro F1-score	0.9026
Test Weighted F1-score	0.9290
low_density Precision	0.9066
low_density Recall	0.9510
low_density F1-score	0.9283
medium_density Precision	0.8600
medium_density Recall	0.7633
medium_density F1-score	0.8088
high_density Precision	0.9639
high_density Recall	0.9776
high_density F1-score	0.9707

TABLE X
DENSITY CLASSIFICATION: CONFUSION MATRIX ON THE TEST SET

True \ Predicted	low	medium	high
low_density	233	11	1
medium_density	23	129	17
high_density	1	10	480

TABLE XI
FIRE/NO-FIRE CLASSIFICATION: FINAL TEST RESULTS

Metric	Result
Test Loss	0.0078
Test Accuracy	99.80%
Test Macro F1-score	0.9922
Test Weighted F1-score	0.9980
fire Precision	0.9855
fire Recall	0.9855
fire F1-score	0.9855
no_fire Precision	0.9989
no_fire Recall	0.9989
no_fire F1-score	0.9989

TABLE XII
FIRE/NO-FIRE CLASSIFICATION: CONFUSION MATRIX ON THE TEST SET

True \ Predicted	fire	no_fire
fire	68	1
no_fire	1	921

validation loss remained close to or lower than the training loss: at the best checkpoint, training loss was 0.0045 while validation loss was 0.0034. The validation accuracy of 99.63% and test accuracy of 99.73% were very close, suggesting stable performance within the prepared dataset distribution.

Several factors may explain the strong test performance. Forest and non-forest images are visually separable in many cases. Forest images typically include tree textures and dense vegetation, while non-forest images include roads, buildings, and bare land. The iterative cleaning process also reduced noisy labels and ambiguous samples, which likely improved class separation. In addition, EfficientNet-B2 with ImageNet pretrained weights provides general visual features that are useful for this task.

TABLE XIII
CROSS-MODULE PERFORMANCE COMPARISON

Module	Classes	Test Acc.	Macro F1
Forest/Non-Forest	2	99.73%	0.9954
Density	3	93.04%	0.9026
Fire/No-Fire	2	99.80%	0.9922

B. Density Classification

The density model achieved 93.04% test accuracy and a macro F1-score of 0.9026. The *low_density* and *high_density* classes were classified with high F1-scores of 0.9283 and 0.9707, respectively, while *medium_density* proved more difficult with an F1-score of 0.8088. The confusion matrix reveals that most errors involve the intermediate *medium_density* class being confused with *low_density* or *high_density*, which is consistent with the visual ambiguity of partial canopy cover and mixed vegetation patterns. The dataset contained 6,027 images across three classes. One reason for the lower *medium_density* performance is likely the dataset size. With only 6,027 images across three classes, the model had less data to learn from compared to the forest/non-forest module. The *medium_density* class is also visually harder to separate from the other two classes. In the next phase of this work, the density dataset can be expanded and the cleaning process can be continued to improve the separation between visually similar density classes.

C. Fire/No-Fire Classification

The fire/no-fire model achieved 99.80% test accuracy, with strong performance on *fire* (F1: 0.9855) and *no_fire* (F1: 0.9989). The confusion matrix shows only two misclassifications among 991 test images, suggesting that the model captured visible fire-related patterns well. However, the *fire* class had far fewer test samples than the *no_fire* class in the test set, with 69 *fire* images compared to 922 *no_fire* images. It should also be noted that the *fire* class contained only 69 images in the test set. With such a small number of samples, the F1-score of 0.9855 should be read carefully, as a few misclassifications would noticeably change the result. A larger and more varied *fire* test set is needed before drawing strong conclusions about the model's reliability.

D. Shared Limitations

Across all three modules, the test sets were drawn from the same data distributions as the training and validation sets. No source-wise, geographically independent, or temporally independent split was applied; therefore, visually similar images may exist across the training, validation, and test sets. As a result, the reported results reflect internal test performance, and generalization to external drone, aerial, or satellite imagery remains unverified.

VI. THREATS TO VALIDITY / LIMITATIONS

- **External validity:** All three test sets were taken from the same data sources as the training and validation sets. This means the results show how well the models perform on familiar data, not on new or unseen imagery. Performance on images from different regions, seasons, sensors, or flight altitudes has not been tested yet, and the current scores should not be treated as a guarantee of real-world performance.
- **Internal validity:** Dataset cleaning for the forest/non-forest module was performed manually, so subjective

judgment during labeling and sample removal may have introduced inconsistencies across iterations. The density and fire/no-fire datasets were also manually reviewed and cleaned but the forest/non-forest dataset had the most extensive iterative cleaning process.

- **Dataset size and class balance:** The density dataset contained 6,027 images across three classes, but the *medium_density* class had fewer samples and lower performance than the *low_density* and *high_density* classes. The fire/no-fire dataset also had class imbalance, especially in the test set, where the *fire* class had substantially fewer samples than the *no_fire* class. These factors may limit the models' ability to learn robust decision boundaries for visually similar or rare cases.
- **Construct validity:** Accuracy and F1-score are useful metrics, but they may not fully capture application-specific requirements for environmental monitoring, such as spatial consistency or robustness under difficult imaging conditions.
- **Conclusion validity:** All experiments were conducted in a Kaggle GPU environment with a fixed seed of 42. Performance may still vary under different hardware settings, software configurations, or random seeds.

VII. CONCLUSION AND FUTURE WORK

This study presented three EfficientNet-B2-based classification models for forest-related image analysis within the Sky Innovators Project. The forest/non-forest model, trained on 34,678 manually labeled and iteratively cleaned images, achieved 99.73% test accuracy and a macro F1-score of 0.9954. The forest density model, trained on 6,027 images across three density levels, achieved 93.04% test accuracy and a macro F1-score of 0.9026. The fire/no-fire model, trained on 6,601 images across two classes, achieved 99.80% test accuracy and a macro F1-score of 0.9922.

These results indicate that EfficientNet-B2 with transfer learning can be effective for forest classification tasks, with performance varying according to the number of classes, dataset size, and the degree of visual overlap between categories. The lower performance of the intermediate *medium_density* class in the density module highlights the challenge of fine-grained forest characterization from image data alone. Because internal test sets were used for all three modules, the results should not be interpreted as universal performance guarantees for external drone, aerial, or satellite imagery.

Future work will focus on:

- Testing all three models on independent external image datasets from different regions, altitudes, seasons, and lighting conditions.
- The density and fire/no-fire datasets will be expanded and the cleaning process will continue to improve dataset quality.

- Comparing EfficientNet-B2 with architectures such as ResNet, MobileNet, DenseNet, and Vision Transformers.
- Applying explainability methods such as Grad-CAM [20], [21] to analyze model decisions across all three tasks and improve the interpretability of forest-related image classification models.

CODE AND DATA AVAILABILITY

The experimental notebooks, datasets, and result reports used in this study are shared in a public GitHub repository to improve transparency and reproducibility.

Repository:

<https://github.com/BurakEfeDeepLearning/Sky-Innovators-Forest-Management>

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